Xiongwei Luo

AI Product & Application Development

Zurich, Switzerland | xiongweiluo1@gmail.com

Internship / Working Student

GitHub / **Portfolio**

EDUCATION

ETH Zurich | MSc Integrated Building Systems *Sep 2025 - Jul 2027 (expected)*

Coursework: Introduction to Machine Learning; Building Agentic AI Systems for Industry Transformation; Scientific Machine and Deep Learning for Design and Construction; Hands-On Deep Learning.

Tongji University | Bachelor of Architecture *Sep 2019 - Jul 2025*

Urban data analysis and human-computer interaction; academic scholarships in four consecutive years.

National University of Singapore | Young Fellowship *Jul 2025*

Summer programme on interdisciplinary AI.

TECHNICAL SKILLS

Programming	Python, SQL
AI and agents	MCP/FastMCP, Function Calling, JSON Schema, QLoRA, Harness evals, paired evaluation
Frameworks and tools	React, FastAPI, Node.js, Docker, Git, GitHub Actions, Figma
Data and systems	MySQL, SQLite, Redis, Celery, WebSocket, automated testing

PROFESSIONAL EXPERIENCE

JD.com | Product Design and Prototyping Intern *Jun 2026 - Sep 2026*

Autonomous Vehicle Product Team

- Built a unified B2B prototype with **17 top-level menus and 91 navigation entries**; integrated 54 entries with existing prototypes and documented requirements for the remaining 37.
- Implemented deep links, history navigation and unsaved-form protection; established **91-page screenshot audits** and interaction checks, and delivered an offline HTML export.

SELECTED PROJECTS

RhinoCoder | Verifiable AI Agent for 3D Modelling

Independent project | GitHub

- Designed a planning, execution and verification loop with **23 versioned MCP tools**, scene inspection and geometric assertions; added cancellation, undo, rollback, idempotency and privacy checks.
- Passed **270/270 real-Rhino runs** on a fixed benchmark of 30 tasks, three configurations and three repetitions; broader generalisation remains unverified.
- QLoRA improved exact tool-call sequences from **20/80 to 77/80** on offline tasks. The real-Rhino quality gate remains unmet; the default production route is unchanged.

LabelHub | AI Annotation and Quality Platform

Team project | Schema architecture, core development & integration | Mock demo

- Built shared schema contracts and a designer-compiler-renderer pipeline; applied preflight checks to field suggestions and Function Calling to AI pre-review, with reviewer diffs for oversight.
- Implemented asynchronous pre-review and exports with retries, idempotency and status tracking; delivered Docker Compose deployment and CI with **nine real-backend E2E scenarios passing**.

Family Garden | Collaborative AI Game

Team project | AI, backend & game systems | Live demo

- Built AI memory and image-analysis workflows with schema validation and moderation; implemented member identity, family data isolation, private photo storage and WebSocket presence.
- Used transactions and operation identifiers for inventory, planting and harvesting; validated the Web release with **26/26 regression checks, 4/4 production E2E scenarios** and iPhone Safari checks.

LANGUAGES

Chinese (native) | English (IELTS 7.0) | German (B1)